

Air University



Network Security (Lab)

Year 2026

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Academic Year: 2024 – 2028

Department of Cyber Security

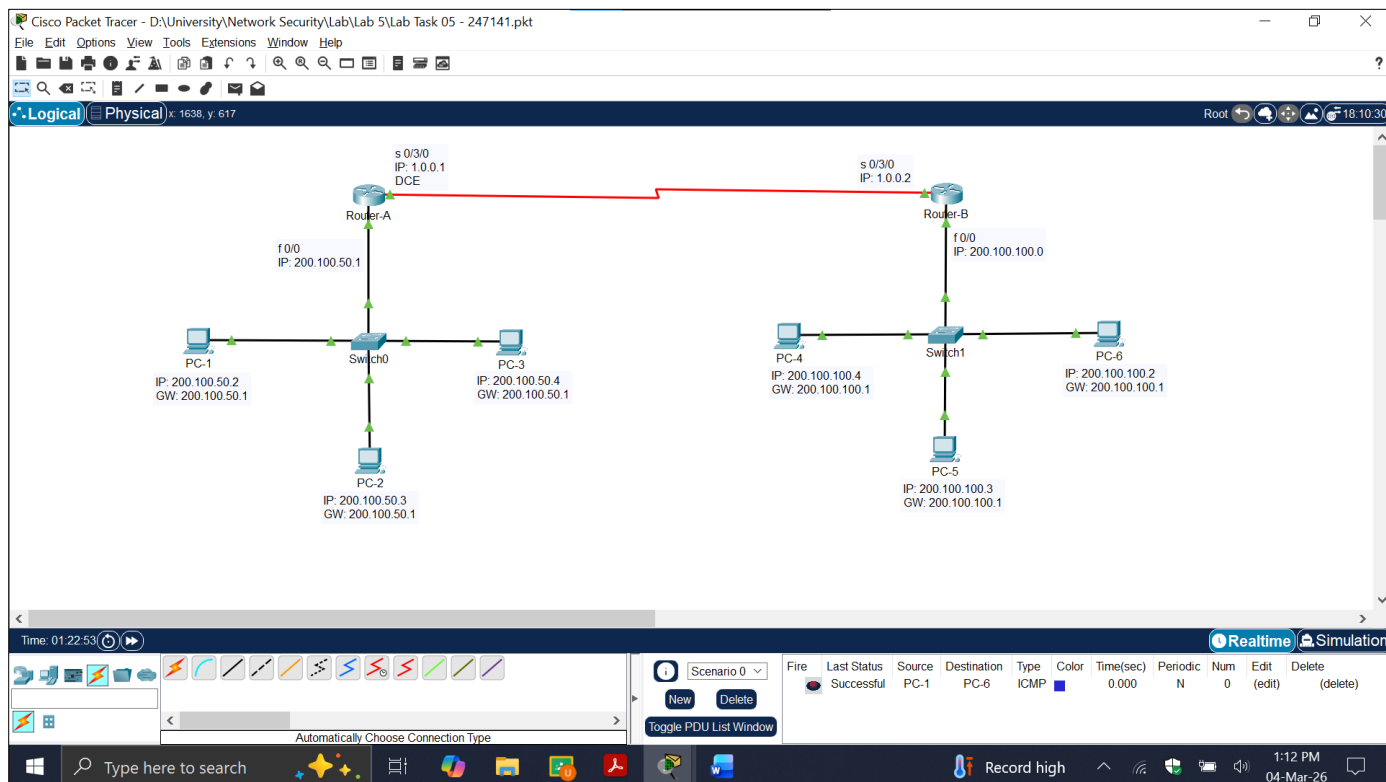
USAMA ARSHAD | 247141

BS – Cyber Security – Fall – 24 (A)

Date of Submission: March 04, 2026

Lab Task # 05

Topology:



1. SSH

Configuring Local User & Verifying:

```
Router-A(config)#username abc secret abc
Router-A(config)#username ali password ali
Router-A(config)#exit
Router-A#
%SYS-5-CONFIG_I: Configured from console by console

Router-A#show run | include username
username abc secret 5 $1$mERr$Dbe7R1LE8b8xdz/DJo0U0
username ali password 0 ali
Router-A#config t
```

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SSH Configuration:

```
Router-A#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-A(config)#ip domain-name riphah.edu.pk
Router-A(config)#crypto key generate rsa
The name for the keys will be: Router-A.riphah.edu.pk
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
```

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Implementation of SSH on Line VTY & Configuring enable Password:

```
Router-A(config)#line vty 0 4
*Mar 1 0:10:48.643: %SSH-5-ENABLED: SSH 1.99 has been enabled
Router-A(config-line)#no password
Router-A(config-line)#local input ssh
^
% Invalid input detected at '^' marker.

Router-A(config-line)#transport input ssh
Router-A(config-line)#login local
Router-A(config-line)#exit
Router-A(config)#enable password Network Security
Router-A(config)#
```

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Accessing Router-A from PC-6:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l abc 1.0.0.1
```

Password:

Router-A>en

Password:

Router-A#



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Accessing Router-A from Router-B through SSH:

```
Router-B>ssh -l abc 1.0.0.1
```

```
Password:
```

```
Router-A>enable
```

```
Password:
```

```
Router-A#
```

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2. ACL

Creation of Access List:

```
Router-A#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-A(config)#access-list 5 deny host 200.100.50.4
Router-A(config)#access-list 5 permit any
Router-A(config)#
```

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Implementation of Access List & Verification:

```
Router-A(config)#int f 0/0
Router-A(config-if)#ip access-group 5 in
Router-A(config-if)#show access-lists
^
% Invalid input detected at '^' marker.

Router-A(config-if)#exit
Router-A(config)#exit
Router-A#
%SYS-5-CONFIG_I: Configured from console by console

Router-A#show ip access-lists
Standard IP access list 5
 10 deny host 200.100.50.4
 20 permit any

Router-A#
```

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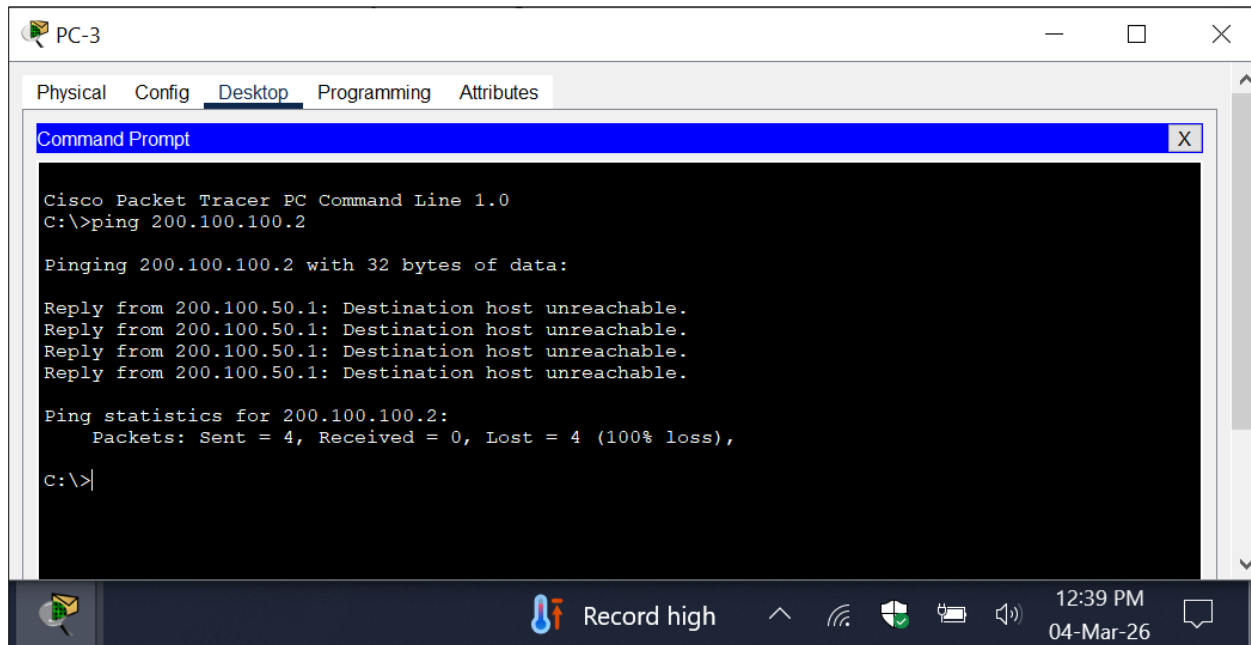
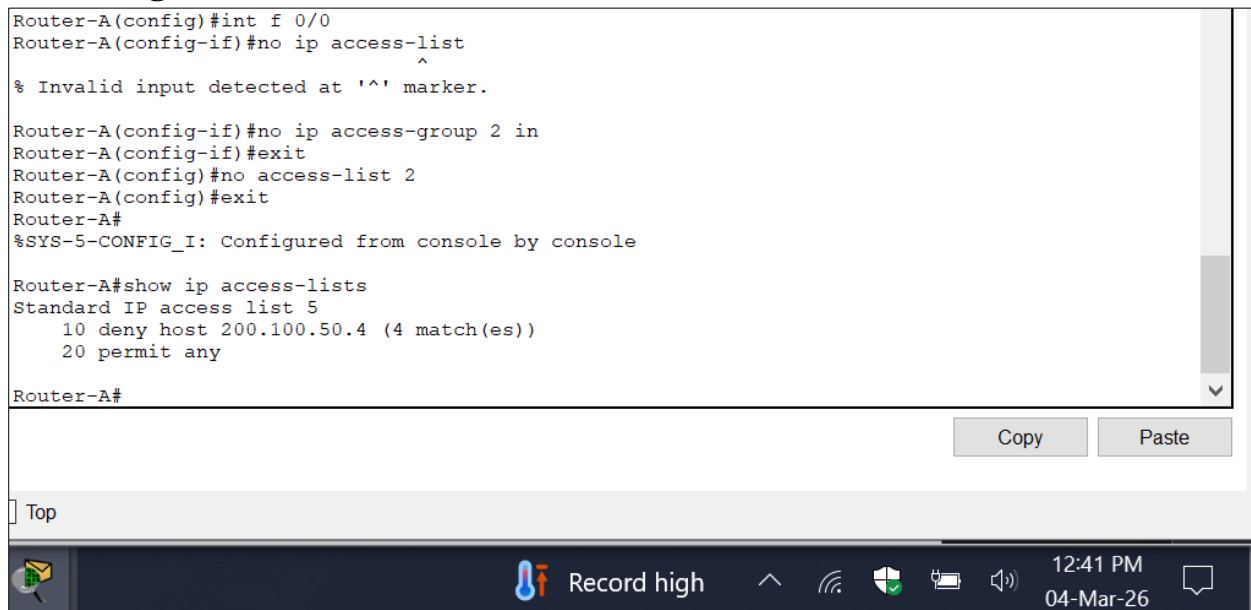
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Ping from PC-3 (ip 200.100.50.4) to PC-6 (ip 200.100.100.2):**Removing Access List:****THE END**